

# SOLIDWORKS API (.NET VBA C#) — Reference (957337): IEndCapFeatureData-IEnumBodies2

SOLIDWORKS API Documentation (converted from CHM)

# IEndCapFeatureData (Type)

## IEndCapFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEndCapFeatureData Interface

Allows access to an end-cap feature.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IEndCapFeatureData
```

Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData
```

C#

```
public interface IEndCapFeatureData
```

C++/CLI

```
public interface class IEndCapFeatureData
```

### Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData](#).

### Example

[Insert Weldment End Cap \(VBA\)](#)  
[Insert Weldment End Cap \(VB.NET\)](#)  
[Insert Weldment End Cap \(C#\)](#)

### Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefinition](#)

### Access Diagram

[EndCapFeatureData](#)

### See Also

[IEndCapFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[IFeatureManager::InsertEndCapFeature2 Method](#)  
[IFeatureManager::InsertEndCapFeature3 Method \(\)](#)

## IEndCapFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEndCapFeatureData Interface

For a list of all members of this type, see [IEndCapFeatureData members](#).

### Public Methods

| Name                                                                                                                 | Description                                                         |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
|  <a href="#">AccessSelections</a> | Gains access to the selections that define this end-cap feature.    |
| <a href="#">ReleaseSelectionAccess</a>                                                                               | Releases access to the selections that define this end-cap feature. |



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## See Also

[IEndCapFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[IFeatureManager::InsertEndCapFeature2 Method](#)  
[IFeatureManager::InsertEndCapFeature3 Method \(\)](#)

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## IEndCapFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEndCapFeatureData Interface

The following tables list the members exposed by [IEndCapFeatureData](#).

## Public Properties

| Name                                                                                                                           | Description                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|  <a href="#">ChamferDistance</a>              | Gets or sets the chamfer distance or fillet radius of the corner of this end-cap feature.                          |
|  <a href="#">DepthDistance</a>                | Gets or sets the depth distance for this end cap feature.                                                          |
|  <a href="#">Face</a>                         | Gets the face that is capped or sets the face to cap.                                                              |
|  <a href="#">IsEndCapInward</a>               | Gets or sets the thickness direction of this end cap feature.                                                      |
|  <a href="#">OffsetDistance</a>               | Gets or sets the offset distance for this end-cap feature.                                                         |
|  <a href="#">Thickness</a>                    | Gets or sets the thickness for this end-cap feature.                                                               |
|  <a href="#">ThicknessRatioForOffset</a>      | Gets or sets the offset distance of the end cap as a ratio of the thickness of the structural member being capped. |
|  <a href="#">UseChamferCorners</a>           | Gets or sets whether to chamfer the corners of this end-cap feature.                                               |
|  <a href="#">UseCornerTreatment</a>         | Gets or sets whether to apply a corner treatment to this end cap feature.                                          |
|  <a href="#">UseReverse</a>                 | Gets or sets whether to reverse the offset of this end cap feature.                                                |
|  <a href="#">UseThicknessRatioForOffset</a> | Gets or sets whether a ratio of the thickness is used to specify the offset for this end-cap feature.              |

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## Public Methods

| Name                                                                                                                       | Description                                                         |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
|  <a href="#">AccessSelections</a>       | Gains access to the selections that define this end-cap feature.    |
|  <a href="#">ReleaseSelectionAccess</a> | Releases access to the selections that define this end-cap feature. |

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## See Also

[IEndCapFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[IFeatureManager::InsertEndCapFeature2 Method](#)  
[IFeatureManager::InsertEndCapFeature3 Method \(\)](#)

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## IEndCapFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEndCapFeatureData Interface

For a list of all members of this type, see [IEndCapFeatureData members](#).

## Public Properties

| Name                                                                                                                | Description                                                                               |
|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|  <a href="#">ChamferDistance</a> | Gets or sets the chamfer distance or fillet radius of the corner of this end-cap feature. |

|                                                                                                                              |                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|  <a href="#">DepthDistance</a>              | Gets or sets the depth distance for this end cap feature.                                                          |
|  <a href="#">Face</a>                       | Gets the face that is capped or sets the face to cap.                                                              |
|  <a href="#">IsEndCapInward</a>             | Gets or sets the thickness direction of this end cap feature.                                                      |
|  <a href="#">OffsetDistance</a>             | Gets or sets the offset distance for this end-cap feature.                                                         |
|  <a href="#">Thickness</a>                  | Gets or sets the thickness for this end-cap feature.                                                               |
|  <a href="#">ThicknessRatioForOffset</a>    | Gets or sets the offset distance of the end cap as a ratio of the thickness of the structural member being capped. |
|  <a href="#">UseChamferCorners</a>          | Gets or sets whether to chamfer the corners of this end-cap feature.                                               |
|  <a href="#">UseCornerTreatment</a>         | Gets or sets whether to apply a corner treatment to this end cap feature.                                          |
|  <a href="#">UseReverse</a>                 | Gets or sets whether to reverse the offset of this end cap feature.                                                |
|  <a href="#">UseThicknessRatioForOffset</a> | Gets or sets whether a ratio of the thickness is used to specify the offset for this end-cap feature.              |

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## See Also

[IEndCapFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[IFeatureManager::InsertEndCapFeature2 Method](#)  
[IFeatureManager::InsertEndCapFeature3 Method \(\)](#)

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## AccessSelections Method (IEndCapFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : AccessSelections Method (IEndCapFeatureData)

Gains access to the selections that define this end-cap feature.

## .NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections( _
    ByVal TopDoc As ModelDoc2, _
    ByVal Component As Component2 _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData
Dim TopDoc As ModelDoc2
Dim Component As Component2
Dim value As System.Boolean
```

```
value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(
    ModelDoc2 TopDoc,
    Component2 Component
)
```

C++/CLI

```
System.bool AccessSelections(
    & ModelDoc2^ TopDoc,
    & Component2^ Component
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*TopDoc*  
 Top-level document (see **Remarks**)  
*Component*

Component in which the feature is to be modified (see **Remarks**)

**Return Value**

True if the selections are successfully accessed, false if not

 **Visual Basic for Applications (VBA) Syntax**

See [EndCapFeatureData::AccessSelections](#).

 **Example**

[List End-cap Feature Properties \(VBA\)](#)

 **Remarks**

| To modify a feature in a... | Then...                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part                        | <ul style="list-style-type: none"><li>TopDoc argument is the <a href="#">IModelDoc2</a> for the part</li><li>Component argument is NULL</li></ul>                                                            |
| Assembly                    | <ul style="list-style-type: none"><li>TopDoc is the IModelDoc2 object for the assembly</li><li>Component argument is the <a href="#">IComponent2</a> object in which the feature is to be modified</li></ul> |

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) or [IFeature::IModifyDefinition2](#) if you modified the feature
- [IEndCapFeatureData::ReleaseSelectionAccess](#) if you did not

 **See Also**

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)

 **Availability**

SOLIDWORKS 2005 FCS, Revision Number 13.0

**ChamferDistance Property (IEndCapFeatureData)**

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : ChamferDistance Property (IEndCapFeatureData)

Gets or sets the chamfer distance or fillet radius of the corner of this end-cap feature.

 **.NET Syntax**

Visual Basic (Declaration)

Property ChamferDistance As System.Double

Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
Dim value As System.Double
```

```
instance.ChamferDistance = value
```

```
value = instance.ChamferDistance
```

C#

```
System.double ChamferDistance {get; set;}
```

C++/CLI

```
property System.double ChamferDistance {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Chamfer distance if [IEndCapFeatureData::UseChamferCorners](#) is true; fillet radius if IEndCapFeatureData::UseChamferCorners is false

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::ChamferDistance](#).

## Example

See the [IEndCapFeatureData](#) examples.

## Remarks

This property is valid only if [IEndCapFeatureData::UseCornerTreatment](#) is true.

## See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)

## Availability

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## DepthDistance Property (IEndCapFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : DepthDistance Property (IEndCapFeatureData)

Gets or sets the depth distance for this end cap feature.

## .NET Syntax

### Visual Basic (Declaration)

Property DepthDistance As System.Double

### Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
Dim value As System.Double
```

```
instance.DepthDistance = value
```

```
value = instance.DepthDistance
```

### C#

```
System.double DepthDistance {get; set;}
```

### C++/CLI

```
property System.double DepthDistance {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Depth distance

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::DepthDistance](#).

## Example

See the [IEndCapFeatureData](#) examples.

## See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)

## Availability

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## Face Property (IEndCapFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : Face Property (IEndCapFeatureData)

Gets the face that is capped or sets the face to cap.

## .NET Syntax

Visual Basic (Declaration)

Property Face As Face2

Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
Dim value As Face2
```

```
instance.Face = value
```

```
value = instance.Face
```

C#

```
Face2 Face {get; set;}
```

C++/CLI

```
property Face2^ Face {  
    Face2^ get();  
    void set ( & Face2^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Pointer to the [IFace2](#) object

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::Face](#).

## Example

[List End-cap Feature Properties \(VBA\)](#)

## Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

## See Also

## Availability

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## IsEndCapInward Property (IEndCapFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : IsEndCapInward Property (IEndCapFeatureData)

Gets or sets the thickness direction of this end cap feature.

## .NET Syntax

### Visual Basic (Declaration)

```
Property IsEndCapInward As System.Integer
```

### Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
Dim value As System.Integer
```

```
instance.IsEndCapInward = value
```

```
value = instance.IsEndCapInward
```

### C#

```
System.int IsEndCapInward {get; set;}
```

### C++/CLI

```
property System.int IsEndCapInward {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Thickness direction as defined by [swEndCapThicknessDirection\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::IsEndCapInward](#).

## Example

See the [IEndCapFeatureData](#) examples.

## See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)

## Availability

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## OffsetDistance Property (IEndCapFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : OffsetDistance Property (IEndCapFeatureData)

Gets or sets the offset distance for this end-cap feature.

## Visual Basic Syntax

### Visual Basic (Declaration)

Property OffsetDistance As System.Double

### Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
Dim value As System.Double
```

```
instance.OffsetDistance = value
```

```
value = instance.OffsetDistance
```

### C#

```
System.double OffsetDistance {get; set;}
```

### C++/CLI

```
property System.double OffsetDistance {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Offset distance

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::OffsetDistance](#).

## Example

See the [IEndCapFeatureData](#) examples.

## Remarks

[IEndCapFeatureData::UseThicknessRatioForOffset](#) must be set to false for this property to have an effect.

## See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)

## Availability

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## ReleaseSelectionAccess Method (IEndCapFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : ReleaseSelectionAccess Method (IEndCapFeatureData)

Releases access to the selections that define this end-cap feature.

## Visual Basic Syntax

### Visual Basic (Declaration)

Sub ReleaseSelectionAccess()

### Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::ReleaseSelectionAccess](#).

### Example

[List End-cap Feature Properties \(VBA\)](#)

### Remarks

[IEndCapFeatureData::AccessSelections](#) puts the model into a rollback state to allow access to the selections that define this feature. Use this method to restore the rollback state if you did not modify the feature or use [IFeature::ModifyDefinition](#) if you did.

### See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)

### Availability

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## Thickness Property (IEndCapFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : Thickness Property (IEndCapFeatureData)

Gets or sets the thickness for this end-cap feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property Thickness As System.Double
```

Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
Dim value As System.Double  
  
instance.Thickness = value  
  
value = instance.Thickness
```

C#

```
System.double Thickness {get; set;}
```

C++/CLI

```
property System.double Thickness {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Thickness

### Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::Thickness](#).

### Example

See the [IEndCapFeatureData](#) examples.

### See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)  
[IEndCapFeatureData::ThicknessRatioForOffset Property](#)  
[IEndCapFeatureData::UseThicknessRatioForOffset Property](#)

### Availability

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## ThicknessRatioForOffset Property (IEndCapFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : ThicknessRatioForOffset Property (IEndCapFeatureData)

Gets or sets the offset distance of the end cap as a ratio of the thickness of the structural member being capped.

### .NET Syntax

Visual Basic (Declaration)

Property ThicknessRatioForOffset As System.Double

Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData
Dim value As System.Double
```

```
instance.ThicknessRatioForOffset = value
```

```
value = instance.ThicknessRatioForOffset
```

C#

```
System.double ThicknessRatioForOffset {get; set;}
```

C++/CLI

```
property System.double ThicknessRatioForOffset {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Ratio of the thickness

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::ThicknessRatioForOffset](#).

## Example

See the [IEndCapFeatureData](#) examples.

## Remarks

[IEndCapFeatureData::UseThicknessRatioForOffset](#) must be set to true for this property to have an effect.

## See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)  
[IEndCapFeatureData::Thickness Property](#)

## Availability

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## UseChamferCorners Property (IEndCapFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : UseChamferCorners Property (IEndCapFeatureData)

Gets or sets whether to chamfer the corners of this end-cap feature.

## .NET Syntax

Visual Basic (Declaration)

```
Property UseChamferCorners As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
Dim value As System.Boolean
```

```
instance.UseChamferCorners = value
```

```
value = instance.UseChamferCorners
```

C#

```
System.bool UseChamferCorners {get; set;}
```

C++/CLI

```
property System.bool UseChamferCorners {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to chamfer corners, false to fillet corners

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::UseChamferCorners](#).

## Example

See the [IEndCapFeatureData](#) examples.

## Remarks

This property is valid only if [IEndCapFeatureData::UseCornerTreatment](#) is true.

Call [IEndCapFeatureData::ChamferDistance](#) to set the chamfer distance or fillet radius of the end cap corner.

## See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)

## Availability

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## UseCornerTreatment Property (IEndCapFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : UseCornerTreatment Property (IEndCapFeatureData)

Gets or sets whether to apply a corner treatment to this end cap feature.

## .NET Syntax

Visual Basic (Declaration)

Property UseCornerTreatment As System.Boolean

Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
Dim value As System.Boolean
```

```
instance.UseCornerTreatment = value
```

```
value = instance.UseCornerTreatment
```

C#

```
System.bool UseCornerTreatment {get; set;}
```

C++/CLI

```
property System.bool UseCornerTreatment {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to apply a corner treatment, false to not

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::UseCornerTreatment](#).

## Example

See the [IEndCapFeatureData](#) examples.

## See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)

## Availability

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## UseReverse Property (IEndCapFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : UseReverse Property (IEndCapFeatureData)

Gets or sets whether to reverse the offset of this end cap feature.

## .NET Syntax

### Visual Basic (Declaration)

```
Property UseReverse As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData  
Dim value As System.Boolean
```

```
instance.UseReverse = value
```

```
value = instance.UseReverse
```

### C#

```
System.bool UseReverse {get; set;}
```

### C++/CLI

```
property System.bool UseReverse {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to reverse the offset, false to not

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::UseReverse](#).

## Example

See the [IEndCapFeatureData](#) examples.

## See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)

## Availability

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## UseThicknessRatioForOffset Property (IEndCapFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEndCapFeatureData Interface](#) : UseThicknessRatioForOffset Property (IEndCapFeatureData)

Gets or sets whether a ratio of the thickness is used to specify the offset for this end-cap feature.

## .NET Syntax

### Visual Basic (Declaration)

Property UseThicknessRatioForOffset As System.Boolean

### Visual Basic (Usage)



```
Dim instance As IEndCapFeatureData
Dim value As System.Boolean

instance.UseThicknessRatioForOffset = value
value = instance.UseThicknessRatioForOffset
```

### C#

```
System.bool UseThicknessRatioForOffset {get; set;}
```

### C++/CLI

```
property System.bool UseThicknessRatioForOffset {
    System.bool get();
    void set ( & System.bool value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to use a ratio of the thickness, false to not

## Visual Basic for Applications (VBA) Syntax

See [EndCapFeatureData::UseThicknessRatioForOffset](#).

## Example

See the [IEndCapFeatureData](#) examples.

## Remarks

| If this property is set to... | Then you can use...                                                                                                                                                      |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| True                          | <a href="#">IEndCapFeatureData::ThicknessRatioForOffset</a> to set the offset distance of the end cap as a ratio of the thickness of the structural member being capped. |
| False                         | <a href="#">IEndCapFeatureData::OffsetDistance</a> to set the offset distance for this end-cap feature.                                                                  |

## See Also

[IEndCapFeatureData Interface](#)  
[IEndCapFeatureData Members](#)  
[IEndCapFeatureData::Thickness Property](#)

## Availability

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# IEntity (Type)

SOLIDWORKS API Help

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## IEntity Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEntity Interface

Allows access to an attribute instance that is stored on an entity.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IEntity
```

Visual Basic (Usage)



```
Dim instance As IEntity
```

C#

```
public interface IEntity
```

C++/CLI

```
public interface class IEntity
```

### Visual Basic for Applications (VBA) Syntax

See [Entity](#).

### Example

[Get Offset Surface Data \(C#\)](#)  
[Get Offset Surface Data \(VB.NET\)](#)  
[Get Offset Surface Data \(VBA\)](#)  
[Select Edges of All Holes on Face \(C#\)](#)  
[Select Edges of All Holes on Face \(VB.NET\)](#)  
[Select Edges of All Holes on Face \(VBA\)](#)

### Remarks

In Microsoft Visual Studio 2008/MFC 9.0 or later, Microsoft added a new interface called IEntity. Your add-in might not compile if it uses both this interface and the SOLIDWORKS IEntity interface.

Possible errors are:

- error C3121: cannot change GUID for class 'IEntity'
- error C2011: 'IEntity' : 'struct' type redefinition
- undefined symbol errors when linking add-ins that do a QueryInterface on IEntity

To avoid errors, you can:

- Include the namespace when declaring an IEntity variable:

```
SldWorks::IEntity ent
```

- Rename IEntity by adding the `rename` function to the `import` of `sldworks.tlb`:

```
#import "sldworks.tlb" rename("IEntity", "SWEntity")
```

and do any of the following:

- use SWEntity instead of IEntity in your code
- do a QueryInterface using the GUID constant of the renamed interface:

```
pFace->QueryInterface(IID_SWEntity, (LPVOID*)&entity);
```

- do a QueryInterface using a GUID lookup of the renamed interface:

```
pFace->QueryInterface(__uuidof(SWEntity), (LPVOID*)&entity);
```

### Accessors

[IAttribute::GetEntity](#)

[IAttribute::IGetEntity2](#)  
[IBrokenOutSectionFeatureData::DepthReference](#)  
[ICoincidentMateFeatureData::EntitiesToMate](#)  
[IComponent2::GetCorrespondingEntity](#) and [IComponent2::IGetCorrespondingEntity](#)  
[IDraftFeatureData2::GetDraftEntities](#) and [IDraftFeatureData2::IGetDraftEntities](#)  
[IEntity::GetSafeEntity](#)  
[IFaultEntity::Entity](#)  
[IHoleSeriesFeatureData2::GetHoleBottomMateEntity](#) and [IHoleSeriesFeatureData2::IGetHoleBottomMateEntity](#)  
[IHoleSeriesFeatureData2::GetHoleTopMateEntity](#) and [IHoleSeriesFeatureData2::IGetHoleTopMateEntity](#)  
[IMateReference::ReferenceEntity](#)  
[IModelDocExtension::GetCorrespondingEntity](#)  
[IMotionPlotAxisFeatureData::Entities](#)  
[IPartDoc::GetEntityByName](#) and [IPartDoc::IGetEntityByName](#)  
[IPartDoc::GetNamedEntities](#) and [IPartDoc::IGetNamedEntities](#)  
[ISurfaceOffsetFeatureData::Entities](#)  
[ISurfaceOffsetFeatureData::IGetEntities](#)  
[IView::GetCorrespondingEntity](#)  
[IView::GetVisibleEntities](#) and [IView::IGetVisibleEntities](#)

 **Access Diagram**

[Entity](#)

 **See Also**

[IEntity Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

**IEntity Interface Members**

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEntity Interface

The following tables list the members exposed by [IEntity](#).

 **Public Properties**

| Name                                                                                                          | Description                                                          |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|  <a href="#">IsSafe</a>    | Gets whether this <a href="#">IEntity</a> object survived a rebuild. |
|  <a href="#">ModelName</a> | Gets or sets the standard Parasolid name attribute of the entity.    |

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 **Public Methods**

| Name                                                                                                                    | Description                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|  <a href="#">FindAttribute</a>       | Finds an attribute on an entity.                                                                                  |
|  <a href="#">GetComponent</a>        | Gets the owning component for this entity.                                                                        |
|  <a href="#">GetDistance</a>         | Gets the minimum or maximum distance between this entity and the given face or edge entity (see <b>Remarks</b> ). |
|  <a href="#">GetDrawingComponent</a> | Gets the drawing component that owns this entity, if the entity is in a drawing view.                             |
|  <a href="#">GetSafeEntity</a>       | Gets a safe entity.                                                                                               |
|  <a href="#">GetType</a>             | Gets the type of the entity.                                                                                      |
|  <a href="#">IFindAttribute</a>      | Finds an attribute on an entity.                                                                                  |
|  <a href="#">IGetComponent</a>       | <b>Obsolete.</b> Superseded by <a href="#">IEntity.IGetComponent2</a> .                                           |
| <a href="#">IGetComponent2</a>                                                                                          | Gets the owning component for this entity.                                                                        |

|                                                                                   |                              |                                                                                                                   |
|-----------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------|
|   |                              |                                                                                                                   |
|  | <a href="#">IGetDistance</a> | Gets the minimum or maximum distance between this entity and the given face or edge entity (see <b>Remarks</b> ). |
|  | <a href="#">Select</a>       | Obsolete. Superseded by <a href="#">IEntity.Select4</a> .                                                         |
|  | <a href="#">Select2</a>      | Obsolete. Superseded by <a href="#">IEntity.Select4</a> .                                                         |
|  | <a href="#">Select3</a>      | Obsolete. Superseded by <a href="#">IEntity.Select4</a> .                                                         |
|  | <a href="#">Select4</a>      | Selects an entity and marks it.                                                                                   |

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## See Also

[IEntity Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

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## IEntity Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEntity Interface

For a list of all members of this type, see [IEntity members](#).

## Public Methods

| Name                                                                                                                   | Description                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|  <a href="#">FindAttribute</a>        | Finds an attribute on an entity.                                                                                  |
|  <a href="#">GetComponent</a>         | Gets the owning component for this entity.                                                                        |
|  <a href="#">GetDistance</a>          | Gets the minimum or maximum distance between this entity and the given face or edge entity (see <b>Remarks</b> ). |
|  <a href="#">GetDrawingComponent</a> | Gets the drawing component that owns this entity, if the entity is in a drawing view.                             |
|  <a href="#">GetSafeEntity</a>      | Gets a safe entity.                                                                                               |
|  <a href="#">GetType</a>            | Gets the type of the entity.                                                                                      |
|  <a href="#">IFindAttribute</a>     | Finds an attribute on an entity.                                                                                  |
|  <a href="#">IGetComponent</a>      | Obsolete. Superseded by <a href="#">IEntity.IGetComponent2</a> .                                                  |
|  <a href="#">IGetComponent2</a>     | Gets the owning component for this entity.                                                                        |
|  <a href="#">IGetDistance</a>       | Gets the minimum or maximum distance between this entity and the given face or edge entity (see <b>Remarks</b> ). |
|  <a href="#">Select</a>             | Obsolete. Superseded by <a href="#">IEntity.Select4</a> .                                                         |
|  <a href="#">Select2</a>            | Obsolete. Superseded by <a href="#">IEntity.Select4</a> .                                                         |
|  <a href="#">Select3</a>            | Obsolete. Superseded by <a href="#">IEntity.Select4</a> .                                                         |
|  <a href="#">Select4</a>            | Selects an entity and marks it.                                                                                   |

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## See Also

[IEntity Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

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## IEntity Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEntity Interface

For a list of all members of this type, see [IEntity members](#).

## Public Properties

| Name                                                                                                       | Description                                                          |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|  <a href="#">IsSafe</a> | Gets whether this <a href="#">IEntity</a> object survived a rebuild. |



[ModelName](#)

Gets or sets the standard Parasolid name attribute of the entity.

[Top](#)

## See Also

[IEntity Interface](#)

[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## FindAttribute Method (IEntity)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : FindAttribute Method (IEntity)

Finds an attribute on an entity.

## .NET Syntax

### Visual Basic (Declaration)

```
Function FindAttribute( _  
    ByVal AttributeDef As System.Object, _  
    ByVal WhichOne As System.Integer _  
) As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IEntity  
Dim AttributeDef As System.Object  
Dim WhichOne As System.Integer  
Dim value As System.Object  
  
value = instance.FindAttribute(AttributeDef, WhichOne)
```

### C#

```
System.object FindAttribute(  
    System.object AttributeDef,  
    System.int WhichOne  
)
```

### C++/CLI

```
System.Object^ FindAttribute(  
& System.Object^ AttributeDef,  
& System.int WhichOne  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*AttributeDef*

Attribute definition you want to find

*WhichOne*

Instance of this type on this entity (0,1,2,3, and so on)

### Return Value

[Attribute](#) instance

## Visual Basic for Applications (VBA) Syntax

See [Entity.FindAttribute](#).

## Example

[Create Attribute \(VBA\)](#)

[Delete Blended Faces \(VBA\)](#)

[Delete Faces \(VBA\)](#)

[Find Attribute \(C#\)](#)

## Remarks

After you add an attribute instance to a face, edge, vertex, loop, feature, or document, you can retrieve the attribute for query or modification in several ways:

- Because an attribute instance is a feature:
  - You can use any of the standard feature traversal functions ([IPartDoc::FeatureByName](#), [IModelDoc2::FirstFeature](#), and [IFeature::GetNextFeature](#)) to get the feature representation of the attribute. You can then use the Feature object that represent the attribute instance with [IFeature::GetSpecificFeature2](#) to get the underlying Attribute object.
  - It can be suppressed. Check its state by using [IAttribute::GetEntityState](#) (swIsEntitySuppressed).
- If the user selected the attribute in the FeatureManager design tree, you can use standard selection control to get the Feature object for the attribute instance ([ISelectionMgr::GetSelectedObject6](#)) and call [Feature::GetSpecificFeature2](#) to get the underlying attribute object.
- If you traverse body topology, you can locate any Attribute object with a call to [Entity::FindAttribute](#) from the Entity object (for example, the face or edge). If the attribute instance was placed on the document, then you must use [Feature::GetSpecificFeature2](#) to get back to it.
- Always call [IAttribute::GetEntityState](#) to check if the attribute instance is valid and unambiguous.

## See Also

[IEntity Interface](#)  
[IEntity Members](#)  
[IEntity::FindAttribute Method](#)  
[IAttribute::GetDefinition Method](#)  
[IAttributeDef::CreateInstance5 Method](#)  
[IBody2::FindAttribute Method](#)  
[IComponent2::FindAttribute Method](#)  
[ISldWorks::DefineAttribute Method](#)

SOLIDWORKS API Help

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## GetComponent Method (IEntity)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : GetComponent Method (IEntity)

Gets the owning component for this entity.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetComponent() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEntity
Dim value As System.Object

value = instance.GetComponent()
```

C#

```
System.object GetComponent()
```

C++/CLI

```
System.Object^ GetComponent();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Component that owns this entity

## Visual Basic for Applications (VBA) Syntax

See [Entity::GetComponent](#).

## Example

[Get Component for Selected Entity \(VBA\)](#)  
[Get Component from Feature \(VBA\)](#)  
[Get Component Via Display Dimension \(VBA\)](#)  
[Get BOM Balloon Properties \(C#\)](#)  
[Get BOM Balloon Properties \(VB.NET\)](#)  
[Get BOM Balloon Properties \(VBA\)](#)

## Remarks

You can use [ISelectionMgr::GetSelectedObject6](#) to get the selected object in the assembly, then use the entity object representing that face or edge object to call this method.

This method works only when you get the [IEntity](#) object directly from the assembly document or from an assembly entity that is displayed in a drawing document.

## See Also

[IEntity Interface](#)  
[IEntity Members](#)  
[IEntity::GetComponent2 Method](#)  
[IAttribute::GetEntity Method](#)  
[IAttribute::GetEntityState Method](#)  
[IComponent2::GetCorrespondingEntity Method](#)  
[IEntity::GetSafeEntity Method](#)  
[IFaultEntity::Entity Property](#)  
[IModelDocExtension::GetCorrespondingEntity Method](#)  
[IPartDoc::GetEntityByName Method](#)  
[IPartDoc::GetNamedEntities Method](#)  
[IView::GetVisibleEntities Method](#)

SOLIDWORKS API Help

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## GetDistance Method (IEntity)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : GetDistance Method (IEntity)

Gets the minimum or maximum distance between this entity and the given face or edge entity (see **Remarks**).

## .NET Syntax

Visual Basic (Declaration)

```
Function GetDistance( _  
    ByVal BaseEntity As System.Object, _  
    ByVal MinDistance As System.Boolean, _  
    ByVal Parameter As System.Object, _  
    ByRef Position1 As System.Object, _  
    ByRef Position2 As System.Object, _  
    ByRef Distance As System.Double _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEntity  
Dim BaseEntity As System.Object  
Dim MinDistance As System.Boolean  
Dim Parameter As System.Object  
Dim Position1 As System.Object  
Dim Position2 As System.Object  
Dim Distance As System.Double  
Dim value As System.Integer
```

```
value = instance.GetDistance(BaseEntity, MinDistance, Parameter, Position1, Position2, Distance)
```

C#

```
System.int GetDistance(  
    System.object BaseEntity,  
    System.bool MinDistance,  
    System.object Parameter,  
    out System.object Position1,
```

```

    out System.object Position2,
    out System.double Distance
)

```

C++/CLI

```

System.int GetDistance(
& System.Object^ BaseEntity,
& System.bool MinDistance,
& System.Object^ Parameter,
& [Out] System.Object^ Position1,
& [Out] System.Object^ Position2,
& [Out] System.double Distance
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*BaseEntity*

[IEntity](#) of another edge or face to get distance from this entity (see **Remarks**)

*MinDistance*

True to return minimum distance between entities; false to return maximum distance between entities

*Parameter*

If `MinDistance = False`,

- and `BaseEntity` is an edge or curve, specify an array of two doubles (Min, Max), where Min and Max make up the U-parameter range of `BaseEntity`
- and `BaseEntity` is a face, specify an array of four doubles (UMin, VMin, UMax, VMax), where UMin, VMin, UMax, and VMax make up the UV-parameter range of the surface of `BaseEntity`

If `MinDistance = True`,

- and `BaseEntity` is an edge or curve, optionally specify an array of two doubles (Min, Max), where Min and Max make up the U-parameter range of `BaseEntity`
- and `BaseEntity` is a face, optionally specify an array of four doubles (UMin, VMin, UMax, VMax), where UMin, VMin, UMax, and VMax make up the UV-parameter range of the surface of `BaseEntity`

*Position1*

Array of doubles; coordinates of this entity that were used to measure the distance

*Position2*

Array of doubles; coordinates of `BaseEntity` that were used to measure the distance

*Distance*

Minimum or maximum distance between entities; if `Parameter` is not specified, minimum distance

#### Return Value

0 if success; -1 if failure

### Visual Basic for Applications (VBA) Syntax

See [Entity::GetDistance](#).

### Example

[Get Distance Between Entities \(VBA\)](#)

[Get Distance Between Entities \(VB.NET\)](#)

[Get Distance Between Entities \(C#\)](#)

### Remarks

Only face and edge entities are supported.

To populate `Parameter` you can find the UV parameter ranges as follows:

- For edges and curves, use [IEdge::GetCurveParams2](#)
- For faces, use [IFace2::GetUVBounds](#)

### See Also

[IEntity Interface](#)

[IEntity Members](#)

[IEntity::GetDistance Method](#)

### Availability

SOLIDWORKS 2009 SP5, Revision Number 17.5

Gets the drawing component that owns this entity, if the entity is in a drawing view.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetDrawingComponent( _  
    ByVal View As View _  
) As DrawingComponent
```

### Visual Basic (Usage)

```
Dim instance As IEntity  
Dim View As View  
Dim value As DrawingComponent  
  
value = instance.GetDrawingComponent(View)
```

### C#

```
DrawingComponent GetDrawingComponent(  
    View View  
)
```

### C++/CLI

```
DrawingComponent^ GetDrawingComponent(  
& View^ View  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*View*

Name of the drawing view in which the entity resides

#### Return Value

[IDrawingComponent](#) object that owns the entity

## Visual Basic for Applications (VBA) Syntax

See [Entity::GetDrawingComponent](#).

## Remarks

If the drawing component is a child component of the referenced assembly, then the drawing component should be a child drawing component in the drawing component configuration in the FeatureManager design tree.

If the entity is in the view, then the drawing component is returned. If the entity is not in the view, then NULL is returned.

## See Also

[IEntity Interface](#)  
[IEntity Members](#)

## Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

## GetSafeEntity Method (IEntity)

[See Also](#) [Example](#)



Gets a safe entity.

## Visual Basic (Declaration)

```
Function GetSafeEntity() As Entity
```

## Visual Basic (Usage)

```
Dim instance As IEntity
Dim value As Entity

value = instance.GetSafeEntity()
```

## C#

```
Entity GetSafeEntity()
```

## C++/CLI

```
Entity^ GetSafeEntity();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Pointer to the [IEntity](#) object

## Visual Basic for Applications (VBA) Syntax

See [Entity::GetSafeEntity](#).

## Example

[Get Faces Affected By Scale Feature \(VBA\)](#)  
[Use Safe Entity \(VBA\)](#)

## Remarks

A safe entity is an entity that continues to be valid after rebuilds until the pointer becomes invalid if the entity object to which it points is deleted. To determine if an entity is safe, use the [IEntity::IsSafe](#) property. This property is read-only and does not persist from session to session.

## See Also

[IEntity Interface](#)  
[IEntity Members](#)  
[IBody2::GetSafeBody Method](#)  
[IBody2::IsSafe Property](#)

## Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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## GetType Method (IEntity)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : GetType Method (IEntity)

Gets the type of the entity.

## Visual Basic (Declaration)

```
Function GetType() As System.Integer
```

## Visual Basic (Usage)



```
Dim instance As IEntity
Dim value As System.Integer

value = instance.GetType()
```

C#

```
System.int GetType()
```

C++/CLI

```
System.int GetType();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Type of entity as defined in [swSelectType\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [Entity::GetType](#).

### Example

[Get Knit Surface Data \(VBA\)](#)  
[Get Mirror Solid Data \(VBA\)](#)  
[Get Component from Feature \(VBA\)](#)  
[Get Component from Feature \(VB.NET\)](#)  
[Get Component from Feature \(C#\)](#)

### Remarks

You can use this method to determine the entity type and the appropriate function calls allowed for the specific type. For example, if this entity is a face, you can call [IFace2::GetArea](#). However, you cannot call [IEdge::GetCurve](#) or [IEdge::IGetCurve](#) because it is on [IEdge](#) object. COM implementations can also use QueryInterface to determine the underlying interface supported by this [IEntity](#) object.

### See Also

[IEntity Interface](#)  
[IEntity Members](#)  
[IAnnotation::GetAttachedEntities2 Method](#)  
[IAttribute::GetEntity Method](#)  
[IAttribute::IGetEntity2 Method](#)  
[IAttribute::GetEntityState Method](#)  
[IComponent2::GetCorrespondingEntity Method](#)  
[IComponent2::IGetCorrespondingEntity Method](#)

### Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

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## IFindAttribute Method (IEntity)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : IFindAttribute Method (IEntity)

Finds an attribute on an entity.

### .NET Syntax

Visual Basic (Declaration)

```
Function IFindAttribute( _  
    ByVal AttributeDef As AttributeDef, _  
    ByVal WhichOne As System.Integer _  
) As Attribute
```

Visual Basic (Usage)



```
Dim instance As IEntity
Dim AttributeDef As AttributeDef
Dim WhichOne As System.Integer
Dim value As Attribute

value = instance.IFindAttribute(AttributeDef, WhichOne)
```

C#

```
Attribute IFindAttribute(
    AttributeDef AttributeDef,
    System.int WhichOne
)
```

C++/CLI

```
Attribute^ IFindAttribute(
    & AttributeDef^ AttributeDef,
    & System.int WhichOne
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*AttributeDef*  
Pointer to the [attribute definition](#) you want to find  
*WhichOne*  
Instance of this type on this entity (0,1,2,3, and so on)

#### Return Value

Pointer to the [attribute](#) instance

### Visual Basic for Applications (VBA) Syntax

See [Entity::IFindAttribute](#).

### Remarks

After you add an attribute instance to a face, edge, vertex, loop, feature, or document, you can retrieve the attribute for query or modification in several ways:

- Because an attribute instance is a feature:
  - You can use any of the standard feature traversal functions ([IPartDoc::IFeatureByName](#), [IModelDoc2::IFirstFeature](#), and [IFeature::IGetNextFeature](#)) to get the feature representation of the attribute. You can then use the Feature object that represents the attribute instance with [IFeature::GetSpecificFeature2](#) to get the underlying Attribute object.
  - It can be suppressed. Check its state by using [IAttribute::GetEntityState](#) (swIsEntitySuppressed).
- If the user selected the attribute in the FeatureManager design tree, you can use standard selection control to get the Feature object for the attribute instance ([ISelectionMgr::GetSelectedObject6](#)) and call [Feature::GetSpecificFeature2](#) to get the underlying attribute object.
- If you traverse body topology, you can locate any Attribute object with a call to [Entity::FindAttribute](#) from the Entity object (for example, the face or edge). If the attribute instance was placed on the document, then you must use [Feature::GetSpecificFeature2](#) to get back to it.
- Always call [IAttribute::GetEntityState](#) to check if the attribute instance is valid and unambiguous.

### See Also

[IEntity Interface](#)  
[Entity Members](#)  
[Entity::FindAttribute Method](#)  
[IAttribute::IGetDefinition Method](#)  
[IAttributeDef::CreateInstance5 Method](#)  
[IBody2::FindAttribute Method](#)  
[IComponent2::IFindAttribute Method](#)  
[ISldWorks::DefineAttribute Method](#)

## IGetComponent Method (IEntity)

[See Also](#)



Obsolete. Superseded by [IEntity::IGetComponent2](#).

 .NET Syntax

Visual Basic (Declaration)

Function IGetComponent() As Component

Visual Basic (Usage)

Dim instance As [IEntity](#)  
Dim value As Component  
  
value = instance.IGetComponent()

C#

Component IGetComponent()

C++/CLI

Component^ IGetComponent();

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [Entity::IGetComponent](#).

 See Also

[IEntity Interface](#)  
[IEntity Members](#)

IGetComponent2 Method (IEntity)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : IGetComponent2 Method (IEntity)

Gets the owning component for this entity.

 .NET Syntax

Visual Basic (Declaration)

Function IGetComponent2() As Component2

Visual Basic (Usage)

Dim instance As [IEntity](#)  
Dim value As Component2  
  
value = instance.IGetComponent2()

C#

Component2 IGetComponent2()

C++/CLI

Component2^ IGetComponent2();

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

**Return Value**

Pointer to the [component](#) that owns this entity

## Visual Basic for Applications (VBA) Syntax

See [Entity::IGetComponent2](#).

## Remarks

You can use [ISelectionMgr::GetSelectedObject6](#) to get the selected object in the assembly, then use the entity object representing that face or edge object to call this method.

This method works only when you get the [IEntity](#) object directly from the assembly document or from an assembly entity that is displayed in a drawing document.

## See Also

[IEntity Interface](#)  
[IEntity Members](#)  
[IEntity::GetComponent Method](#)  
[IAnnotation::GetAttachedEntities2 Method](#)  
[IAttribute::IGetEntity2 Method](#)  
[IAttribute::GetEntityState Method](#)  
[IComponent2::IGetCorrespondingEntity Method](#)  
[IEntity::GetSafeEntity Method](#)  
[IFaultEntity::Entity Property](#)  
[IModelDocExtension::GetCorrespondingEntity Method](#)  
[IPartDoc::IGetEntityName Method](#)  
[IPartDoc::IGetNamedEntities Method](#)  
[IView::IGetVisibleEntities Method](#)

SOLIDWORKS API Help

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## IGetDistance Method (IEntity)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : IGetDistance Method (IEntity)

Gets the minimum or maximum distance between this entity and the given face or edge entity (see **Remarks**).

## .NET Syntax

Visual Basic (Declaration)

```
Function IGetDistance( _  
    ByVal BaseEntity As IEntity, _  
    ByVal MinDistance As System.Boolean, _  
    ByRef Parameter As System.Double, _  
    ByRef Position1 As System.Double, _  
    ByRef Position2 As System.Double, _  
    ByRef Distance As System.Double _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEntity  
Dim BaseEntity As IEntity  
Dim MinDistance As System.Boolean  
Dim Parameter As System.Double  
Dim Position1 As System.Double  
Dim Position2 As System.Double  
Dim Distance As System.Double  
Dim value As System.Integer  
  
value = instance.IGetDistance(BaseEntity, MinDistance, Parameter, Position1, Position2, Distance)
```

C#

```

System.int IGetDistance(
    Entity BaseEntity,
    System.bool MinDistance,
    ref System.double Parameter,
    out System.double Position1,
    out System.double Position2,
    out System.double Distance
)

```

C++/CLI

```

System.int IGetDistance(
    & Entity^ BaseEntity,
    & System.bool MinDistance,
    & System.double% Parameter,
    & [Out] System.double Position1,
    & [Out] System.double Position2,
    & [Out] System.double Distance
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*BaseEntity*

[IEntity](#) of another edge or face to get distance from this entity (see **Remarks**)

*MinDistance*

True to return minimum distance; false to return maximum distance

*Parameter*

For in-process, unmanaged C++:

If *MinDistance* = False,

- and *BaseEntity* is an edge or curve, specify an array of two doubles (Min, Max), where Min and Max make up the U-parameter range of *BaseEntity*
- and *BaseEntity* is a face, specify an array of four doubles (UMin, VMin, UMax, VMax), where UMin, VMin, UMax, and VMax make up the UV-parameter range of the surface of *BaseEntity*

If *MinDistance* = True,

- and *BaseEntity* is an edge or curve, optionally specify an array of two doubles (Min, Max), where Min and Max make up the U-parameter range of *BaseEntity*
- and *BaseEntity* is a face, optionally specify an array of four doubles (UMin, VMin, UMax, VMax), where UMin, VMin, UMax, and VMax make up the UV-parameter range of the surface of *BaseEntity*

For VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

*Position1*

- For in-process, unmanaged C++: pointer to an array of doubles (coordinates of this entity that were used to measure the distance)
- For VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

*Position2*

- For in-process, unmanaged C++: pointer to an array of doubles (coordinates of *BaseEntity* that were used to measure the distance)

- For VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

*Distance*

Minimum or maximum distance between entities; if *Parameter* is not specified, the minimum distance

#### Return Value

0 if success; -1 if failure

#### Remarks

Only face and edge entities are supported.

To populate *Parameter* you can find the UV parameter ranges as follows:

- For edges and curves, use [IEdge::GetCurveParams2](#)
- For faces, use [IFace2::GetUVBounds](#)

#### See Also

[IEntity Interface](#)  
[IEntity Members](#)  
[IEntity::GetDistance Method](#)

#### Availability

## IsSafe Property (IEntity)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : IsSafe Property (IEntity)

Gets whether this [IEntity](#) object survived a rebuild.

### .NET Syntax

#### Visual Basic (Declaration)

```
ReadOnly Property IsSafe As System.Boolean
```

#### Visual Basic (Usage)



```
Dim instance As IEntity  
Dim value As System.Boolean  
  
value = instance.IsSafe
```

#### C#

```
System.bool IsSafe {get;}
```

#### C++/CLI

```
property System.bool IsSafe {  
    System.bool get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True if the [IEntity](#) object survived the rebuild, false if not

### Visual Basic for Applications (VBA) Syntax

See [Entity::IsSafe](#).

### Remarks

The [IEntity::IsSafe](#) property is read-only and does not persist from session to session. To make an entity safe, use [IEntity::GetSafeEntity](#).

### See Also

[IEntity Interface](#)  
[IEntity Members](#)

### Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

## ModelName Property (IEntity)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : ModelName Property (IEntity)

Gets or sets the standard Parasolid name attribute of the entity.

### .NET Syntax

#### Visual Basic (Declaration)

```
Property ModelName As System.String
```

#### Visual Basic (Usage)



```
Dim instance As IEntity
Dim value As System.String

instance.ModelName = value

value = instance.ModelName
```

C#

```
System.string ModelName {get; set;}
```

C++/CLI

```
property System.String^ ModelName {
    System.String^ get();
    void set ( & System.String^ value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Standard Parasolid name attribute

### Visual Basic for Applications (VBA) Syntax

See [Entity::ModelName](#).

### Example

This Visual Basic subroutine shows how to write a Parasolid name attribute to each face in the model.

```
Private Sub NameAllFaces (ByRef swBody as SldWorks.body2)
```

```
    Dim swFace as Sldworks.Face2
    Dim swEnt As Sldworks.Entity
    Dim Name, RootName As String
    Dim Index As Integer
    Dim ret as Boolean

    swFace = swBody.GetFirstFace

    Index = 0
    RootName = "My Face #"

    Do While Not swFace is Nothing
        swEnt = swFace
        Name = RootName + str(Index)
        swEnt.ModelName = Name
        Index = Index + 1
        swFace = swFace.GetNextFace
    Loop
```

```
End Sub
```

### See Also

## Select Method (IEntity)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : Select Method (IEntity)

Obsolete. Superseded by [IEntity::Select4](#).

### .NET Syntax

#### Visual Basic (Declaration)

```
Function Select( _  
    ByVal AppendFlag As System.Boolean _  
) As System.Boolean
```

#### Visual Basic (Usage)



```
Dim instance As IEntity  
Dim AppendFlag As System.Boolean  
Dim value As System.Boolean  
  
value = instance.Select(AppendFlag)
```

#### C#

```
System.bool Select(  
    System.bool AppendFlag  
)
```

#### C++/CLI

```
System.bool Select(  
&    System.bool AppendFlag  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*AppendFlag*

### Visual Basic for Applications (VBA) Syntax

See [Entity::Select](#).

### See Also

[IEntity Interface](#)  
[IEntity Members](#)

## Select2 Method (IEntity)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : Select2 Method (IEntity)

Obsolete. Superseded by [IEntity::Select4](#).

### .NET Syntax

#### Visual Basic (Declaration)

```
Function Select2( _  
    ByVal Append As System.Boolean, _  
    ByVal Mark As System.Integer _  
) As System.Boolean
```

```
Dim instance As IEntity
Dim Append As System.Boolean
Dim Mark As System.Integer
Dim value As System.Boolean

value = instance.Select2(Append, Mark)
```

C#

```
System.bool Select2(
    System.bool Append,
    System.int Mark
)
```

C++/CLI

```
System.bool Select2(
    & System.bool Append,
    & System.int Mark
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Append*

*Mark*

### Visual Basic for Applications (VBA) Syntax

See [Entity::Select2](#).

### See Also

[IEntity Interface](#)  
[IEntity Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## Select3 Method (IEntity)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : Select3 Method (IEntity)

Obsolete. Superseded by [IEntity::Select4](#).

### .NET Syntax

Visual Basic (Declaration)

```
Function Select3( _
    ByVal Append As System.Boolean, _
    ByVal Mark As System.Integer, _
    ByVal Callout As Callout _
) As System.Boolean
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IEntity
Dim Append As System.Boolean
Dim Mark As System.Integer
Dim Callout As Callout
Dim value As System.Boolean

value = instance.Select3(Append, Mark, Callout)
```

C#

```
System.bool Select3(
    System.bool Append,
    System.int Mark,
    Callout Callout
)
```

C++/CLI

```
System.bool Select3(
    & System.bool Append,
    & System.int Mark,
    & Callout^ Callout
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Append*

*Mark*

*Callout*

### Visual Basic for Applications (VBA) Syntax

See [Entity::Select3](#).

### See Also

[IEntity Interface](#)  
[IEntity Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## Select4 Method (IEntity)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEntity Interface](#) : Select4 Method (IEntity)

Selects an entity and marks it.

### .NET Syntax

Visual Basic (Declaration)

```
Function Select4( _
    ByVal Append As System.Boolean, _
    ByVal Data As SelectData _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IEntity
Dim Append As System.Boolean
Dim Data As SelectData
Dim value As System.Boolean

value = instance.Select4(Append, Data)
```

C#

```
System.bool Select4(
    System.bool Append,
    SelectData Data
)
```

C++/CLI

```
System.bool Select4(
    & System.bool Append,
```



```
& SelectData^ Data  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

##### *Append*

True appends the entity to the selection list, false replaces the selection list with this entity

##### *Data*

Pointer to the [ISelectData](#) object

#### Return Value

True if the selection is successful, false if not

### Visual Basic for Applications (VBA) Syntax

See [Entity::Select4](#).

### Example

[Check Interference Using IAssemblyDoc::ToolsCheckInterference2 \(VBA\)](#)  
[Get Faces Associated with Feature \(VBA\)](#)  
[Get Reference Point Data \(VBA\)](#)  
[Get Visible Components and Entities in Drawing View \(VBA\)](#)  
[Select Loop of Edges \(VBA\)](#)  
[Use Safe Entity \(VBA\)](#)  
[Get Offset Surface Data \(C#\)](#)  
[Get Offset Surface Data \(VB.NET\)](#)  
[Get Offset Surface Data \(VBA\)](#)  
[Select Edges of All Holes on Face \(C#\)](#)  
[Select Edges of All Holes on Face \(VB.NET\)](#)  
[Select Edges of All Holes on Face \(VBA\)](#)

### Remarks

When you use this method, selection behaves differently depending on the command state of SOLIDWORKS. One case is when SOLIDWORKS is running a command that has a PropertyManager associated with a selection list box (for example, a feature creation command such as **Insert Fillet**). The selection behavior in this case is:

- Selecting a new entity appends it to the selection list.
  - or -
- Selecting an entity that is already selected deselects the entity.

SOLIDWORKS ignores the Append argument because the selection is always appended to the selection list.

The second case is when there is no command running, which is the default state of SOLIDWORKS.

You can use this method only with [IEntity](#) objects that you get from the active document. For example, if Assembly1 is the active document when you call this method, then you must get the entity directly from the Assembly1 document. You can do this using items in the selection list (for example, [ISelectionMgr::GetSelectedObject6](#)) or you can traverse the body of an assembly component (for example, [IComponent2::GetBody](#) and [IBody2::GetFirstFace](#)). You cannot obtain the entity from the underlying part document (for example, [IComponent2::GetModelDoc](#) or [IBody2::GetFirstFace](#)).

You can use this method to select features. In Visual C++, you must first get the [IEntity](#) interface from the [IFeature](#) object.

### See Also

[IEntity Interface](#)  
[IEntity Members](#)

### Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

# IEnumBodies (Type)

## IEnumBodies Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEnumBodies Interface

Obsolete. Superseded by [IEnumBodies2](#).

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IEnumBodies
```

Visual Basic (Usage)



```
Dim instance As IEnumBodies
```

C#

```
public interface IEnumBodies
```

C++/CLI

```
public interface class IEnumBodies
```

### Visual Basic for Applications (VBA) Syntax

See [EnumBodies](#).

### See Also

[IEnumBodies Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## IEnumBodies Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEnumBodies Interface

For a list of all members of this type, see [IEnumBodies members](#).

### Public Methods

| Name                  | Description                                                  |
|-----------------------|--------------------------------------------------------------|
| <a href="#">Clone</a> | Obsolete. Superseded by <a href="#">IEnumBodies2.Clone</a> . |
| <a href="#">Next</a>  | Obsolete. Superseded by <a href="#">IEnumBodies2.Next</a> .  |
| <a href="#">Reset</a> | Obsolete. Superseded by <a href="#">IEnumBodies2.Reset</a> . |
| <a href="#">Skip</a>  | Obsolete. Superseded by <a href="#">IEnumBodies2.Skip</a> .  |

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### See Also

[IEnumBodies Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## IEnumBodies Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEnumBodies Interface

The following tables list the members exposed by [IEnumBodies](#).

| Public Methods                                                                                          |                                                              |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Name                                                                                                    | Description                                                  |
|  <a href="#">Clone</a> | Obsolete. Superseded by <a href="#">IEnumBodies2.Clone</a> . |
|  <a href="#">Next</a>  | Obsolete. Superseded by <a href="#">IEnumBodies2.Next</a> .  |
|  <a href="#">Reset</a> | Obsolete. Superseded by <a href="#">IEnumBodies2.Reset</a> . |
|  <a href="#">Skip</a>  | Obsolete. Superseded by <a href="#">IEnumBodies2.Skip</a> .  |

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See Also

[IEnumBodies Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

### Clone Method (IEnumBodies)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnumBodies Interface](#) : Clone Method (IEnumBodies)

Obsolete. Superseded by [IEnumBodies2.Clone](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub Clone( _  
    ByRef Ppenum As EnumBodies _  
)
```

Visual Basic (Usage)



```
Dim instance As IEnumBodies  
Dim Ppenum As EnumBodies  
  
instance.Clone(Ppenum)
```

C#

```
void Clone(  
    out EnumBodies Ppenum  
)
```

C++/CLI

```
void Clone(  
& [Out] EnumBodies^ Ppenum  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

*Ppenum*

Visual Basic for Applications (VBA) Syntax

See [EnumBodies.Clone](#).

See Also

[IEnumBodies Interface](#)  
[IEnumBodies Members](#)

## Next Method (IEnumBodies)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnumBodies Interface](#) : Next Method (IEnumBodies)

Obsolete. Superseded by [IEnumBodies2::Next](#).

### .NET Syntax

#### Visual Basic (Declaration)

```
Sub Next( _  
    ByVal Celt As System.Integer, _  
    ByRef Rgelt As Body, _  
    ByRef PceltFetched As System.Integer _  
)
```

#### Visual Basic (Usage)



```
Dim instance As IEnumBodies  
Dim Celt As System.Integer  
Dim Rgelt As Body  
Dim PceltFetched As System.Integer  
  
instance.Next(Celt, Rgelt, PceltFetched)
```

#### C#

```
void Next(  
    System.int Celt,  
    out Body Rgelt,  
    out System.int PceltFetched  
)
```

#### C++/CLI

```
void Next(  
& System.int Celt,  
& [Out] Body^ Rgelt,  
& [Out] System.int PceltFetched  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Celt*

*Rgelt*

*PceltFetched*

### Visual Basic for Applications (VBA) Syntax

See [EnumBodies::Next](#).

### See Also

[IEnumBodies Interface](#)  
[IEnumBodies Members](#)

## Reset Method (IEnumBodies)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnumBodies Interface](#) : Reset Method (IEnumBodies)

Obsolete. Superseded by [IEnumBodies2::Reset](#).

## .NET Syntax

Visual Basic (Declaration)

```
Sub Reset()
```

Visual Basic (Usage)



```
Dim instance As IEnumBodies  
  
instance.Reset()
```

C#

```
void Reset()
```

C++/CLI

```
void Reset();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Visual Basic for Applications (VBA) Syntax

See [EnumBodies::Reset](#).

## See Also

[IEnumBodies Interface](#)  
[IEnumBodies Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

### Skip Method (IEnumBodies)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnumBodies Interface](#) : Skip Method (IEnumBodies)

Obsolete. Superseded by [IEnumBodies2::Skip](#).

## .NET Syntax

Visual Basic (Declaration)

```
Sub Skip(  
    ByVal Celt As System.Integer  
)
```

Visual Basic (Usage)



```
Dim instance As IEnumBodies  
Dim Celt As System.Integer  
  
instance.Skip(Celt)
```

C#

```
void Skip(  
    System.int Celt  
)
```

C++/CLI

```
void Skip(  
& System.int Celt  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

## Visual Basic for Applications (VBA) Syntax

See [EnumBodies::Skip](#).

## See Also

[IEnumBodies Interface](#)  
[IEnumBodies Members](#)

# IEnumBodies2 (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## IEnumBodies2 Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEnumBodies2 Interface

Allows access to [bodies](#) enumeration.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IEnumBodies2
```

Visual Basic (Usage)

```
Dim instance As IEnumBodies2
```

C#

```
public interface IEnumBodies2
```

C++/CLI

```
public interface class IEnumBodies2
```

### Visual Basic for Applications (VBA) Syntax

See [EnumBodies2](#).

### Example

[Enumerate Bodies \(C++\)](#)

### Remarks

For use in in-process DLLs only.

### Accessors

[IBody2::MatchedBoolean3](#) and [IBody2::IMatchedBoolean3](#)  
[IBody2::Operations2](#) and [IBody2::IOperations2](#)  
[IComponent2::EnumBodies2](#)  
[IComponent2::EnumRelatedBodies](#)  
[IComponent2::EnumSectionedBodies](#)  
[IEnumBodies2::Clone](#)  
[IPartDoc::EnumBodies3](#)  
[IPartDoc::EnumRelatedBodies2](#)  
[IPartDoc::EnumRelatedSectionedBodies2](#)

### Access Diagram

[EnumBodies2](#)

### See Also

[IEnumBodies2 Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## IEnumBodies2 Interface Methods

[See Also](#)

SolidWorks.Interop.sldworks Namespace : IEnumBodies2 Interface

For a list of all members of this type, see [IEnumBodies2 members](#).

### Public Methods

| Name                                                                                                    | Description                                                     |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
|  <a href="#">Clone</a> | Clones a bodies enumeration.                                    |
|  <a href="#">Next</a>  | Gets the next body in the bodies enumeration.                   |
|  <a href="#">Reset</a> | Resets the sequence to the beginning of the bodies enumeration. |
|  <a href="#">Skip</a>  | Skips the specified number of bodies in the bodies enumeration. |

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### See Also

[IEnumBodies2 Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## IEnumBodies2 Interface Members

[See Also](#) [Methods](#)

SolidWorks.Interop.sldworks Namespace : IEnumBodies2 Interface

The following tables list the members exposed by [IEnumBodies2](#).

### Public Methods

| Name                                                                                                      | Description                                                     |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
|  <a href="#">Clone</a> | Clones a bodies enumeration.                                    |
|  <a href="#">Next</a>  | Gets the next body in the bodies enumeration.                   |
|  <a href="#">Reset</a> | Resets the sequence to the beginning of the bodies enumeration. |
|  <a href="#">Skip</a>  | Skips the specified number of bodies in the bodies enumeration. |

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### See Also

[IEnumBodies2 Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## Clone Method (IEnumBodies2)

[See Also](#)

SolidWorks.Interop.sldworks Namespace > [IEnumBodies2 Interface](#) : Clone Method (IEnumBodies2)

Clones a bodies enumeration.

### .NET Syntax

Visual Basic (Declaration)

```
Sub Clone( _  
    ByRef Ppenum As EnumBodies2 _  
)
```

Visual Basic (Usage)



```
Dim instance As IEnumBodies2  
Dim Ppenum As EnumBodies2
```

```
instance.Clone(Ppenum)
```

C#

```
void Clone(  
    out EnumBodies2 Ppenum  
)
```

C++/CLI

```
void Clone(  
& [Out] EnumBodies2^ Ppenum  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Ppenum*

Pointer to the [bodies enumeration](#)

### Visual Basic for Applications (VBA) Syntax

See [EnumBodies2::Clone](#).

### Remarks

For use in in-process DLLs only.

### See Also

[IEnumBodies2 Interface](#)  
[IEnumBodies2 Members](#)

### Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## Next Method (IEnumBodies2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnumBodies2 Interface](#) : Next Method (IEnumBodies2)

Gets the next body in the bodies enumeration.

### .NET Syntax

Visual Basic (Declaration)

```
Sub Next( _  
    ByVal Celt As System.Integer, _  
    ByRef Rgelt As Body2, _  
    ByRef PceltFetched As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IEnumBodies2  
Dim Celt As System.Integer  
Dim Rgelt As Body2  
Dim PceltFetched As System.Integer  
  
instance.Next(Celt, Rgelt, PceltFetched)
```

C#

```
void Next(
    System.int Celt,
    out Body2 Rgelt,
    out System.int PceltFetched
)
```

C++/CLI

```
void Next(
    & System.int Celt,
    & [Out] Body2^ Rgelt,
    & [Out] System.int PceltFetched
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Celt*

Number of bodies for the bodies enumeration

*Rgelt*

Pointer to an array of [bodies](#) of size Celt

*PceltFetched*

Pointer to the number of bodies returned from the list; this value can be less than Celt if you request more bodies than exist, or it can be NULL if no more bodies exist

### Visual Basic for Applications (VBA) Syntax

See [EnumBodies2::Next](#).

### Example

[Enumerate Bodies \(C++\)](#)

### Remarks

For use in in-process DLLs only.

### See Also

[IEnumBodies2 Interface](#)  
[IEnumBodies2 Members](#)

### Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## Reset Method (IEnumBodies2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnumBodies2 Interface](#) : Reset Method (IEnumBodies2)

Resets the sequence to the beginning of the bodies enumeration.

### .NET Syntax

Visual Basic (Declaration)

```
Sub Reset()
```

Visual Basic (Usage)



```
Dim instance As IEnumBodies2
```

```
instance.Reset()
```

C#

```
void Reset()
```

C++/CLI

```
void Reset();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Visual Basic for Applications (VBA) Syntax

See [EnumBodies2::Reset](#).

## Remarks

For use in in-process DLLs only.

## See Also

[IEnumBodies2 Interface](#)  
[IEnumBodies2 Members](#)

## Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic](#).

## Skip Method (IEnumBodies2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnumBodies2 Interface](#) : Skip Method (IEnumBodies2)

Skips the specified number of bodies in the bodies enumeration.

## .NET Syntax

Visual Basic (Declaration)

```
Sub Skip(  
    ByVal Celt As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IEnumBodies2  
Dim Celt As System.Integer  
  
instance.Skip(Celt)
```

C#

```
void Skip(  
    System.int Celt  
)
```

C++/CLI

```
void Skip(  
    & System.int Celt  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*Celt*  
Number of bodies to skip

## Visual Basic for Applications (VBA) Syntax

See [EnumBodies2::Skip](#).

## Remarks

For use in in-process DLLs only.

## See Also

[IEnumBodies2 Interface](#)  
[IEnumBodies2 Members](#)

## Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0